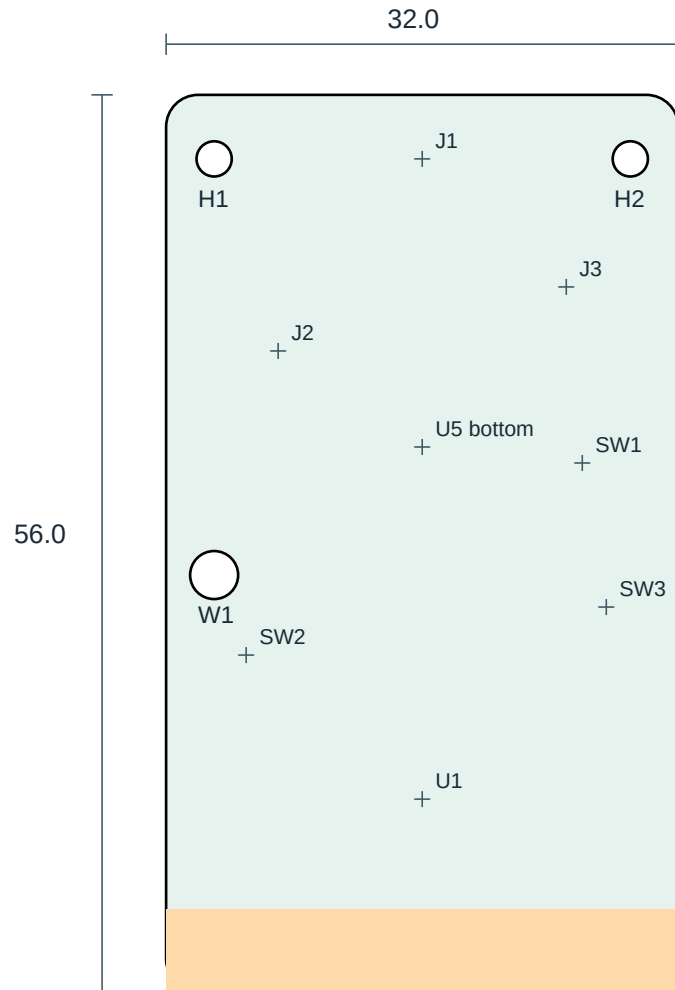


MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R2

PCB outline and placement datum - retained review-board geometry



PCB: 32 x 56 x 1.0, two layers, FR4, corners R2.0 [A]

Outline tolerance target ± 0.2 ; thickness ± 0.1 [A]

Datum: PCB centre XY; +Y toward USB; +Z above PCB.

Mount H1/H2: (-13, +24) and (+13, +24), dia 2.2 NPTH [A]

Harness W1: (-13, -2), dia 3.0 NPTH [A]

J1: (0,24), rotation 180 deg, top [A,B]

J2: (-9,12), entry toward -Y, lid removed to mate [A,C]

J3: (9,16), entry from +Z, lid removed to program [A,D]

Shutter SW2 (-11,-7), pair SW3 (11.5,-4), top [A,E]

Power SW1 (10,5), rotation 90 deg; slide travel 1.6 [A,F]

U1 (0,-16); antenna toward -Y [A,G]

U5 temperature sensor: (0,6), underside [A,H]

Antenna exclusion: full width, Y=-28 to -22.9 [A,G]

No copper, battery, wire, fastener or other part in strip.

Crosses mark anchors, not footprint drawings.

[A] Original engineering coordinates, src/remote-circuit.tsx

[B] GCT-USB4105-120.pdf; current J1 drill gap FAILS 0.2 target.

[C,D] JST-PH.pdf / JST-SH.pdf official dimensioned drawings.

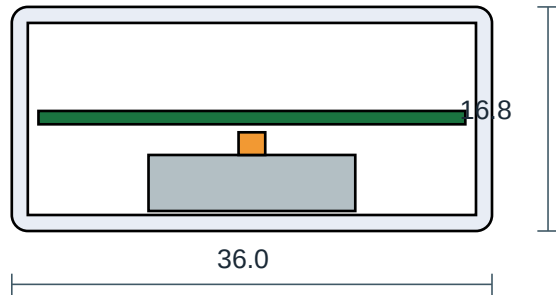
[E,F] C720477-manufacturer.pdf / SHOUHAN-MSK12C02.pdf

[G] Ebyte E73 manual + radio-land-audit.json

[H] TI TMP390.pdf; body/land dimensions, DRL package.

MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R2

Remote enclosure and battery stack - assembly datums



Section: Z=0 is external enclosure back [A]

Base floor 1.2; PCB bottom 8.0; PCB top 9.0 [A]

Lid underside 15.6; external lid top 16.8 [A]

Battery top assembly datum 5.7 +/-0.1 [A,I]

Thermal contact at (0,6): 2 x 2, gap 1.7 nominal [A,H,J]

Bottom C8: maximum height 1.45; nominal cell gap 0.85 [K]

PH mated top Z=14.5; lid clearance 1.1 nominal [C]

SH mated top Z=15.3; programming with lid removed [D]

Remote body 36 x 60 x 16.8, corners R3 [A]

Maximum width with slider 38.4; cap top Z=17.5 [A]

Internal cavity 33.6 x 57.6; walls 1.2 [A]

PCB-wall gap 0.8 nominal; calculated minimum 0.5 [A]

Printed dimensional tolerance target +/-0.2 [A]

Pack: DATA POWER DTP401525(PHR), external [I]

Maximum 15.5 x 27 x 4.2; nest 16.5 x 28 [A,I]

Pack plan: X +/-7.75, Y=-4.5 to +22.5 [A]

Measure actual thickness; add insulating bed shims.

Do not compress pouch to establish Z datum.

Thermal pad free 2.032; nominal compression 16.3% [J]

USB opening 12 x 7, centre (X=0,Y=29,Z=11) [A,B]

This opening must be rechecked for a replacement J1.

Two M2 x 14 lid screws; dia 1.7 pilot / 2.4 clearance [A]

Board support boss dia 4.2, top Z=8; lid boss Z=9.15 [A]

[I] DTP401525-PHR-v2.pdf, pages 4, 7, 9.

[J] Henkel BERGQUIST-GAP-PAD-TGP-1500.pdf

GP1500-0.080-02-0404 sheet, cut 2 x 2.

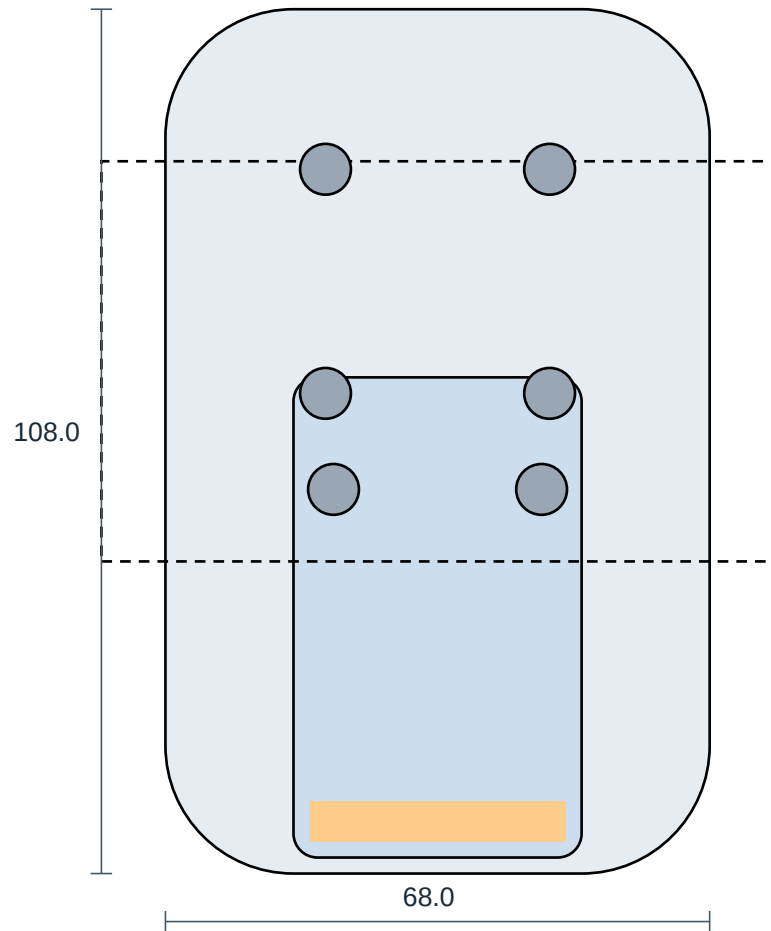
[K] Samsung CL21A106KAYNNNE package height.

All [A] dimensions are original design allocations.

Thermal gradient, pad pressure, swelling and fit are untested.

MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R2

Grip, docking, magnets and antenna clearance



Grip plate: 68 x 108 x 6.5, R16, centre Y=-20 [A]

Remote dock transform: Y=-42, back Z=6.8 [A]

Cradle inside width 36.6; nominal side gap 0.3 [A]

Rails 1.6 x 48 x 3.3; centres X=±19.1, Y=-38 [A]

End stop 36.6 x 1.6 x 1.2; centre Y=-12 [A]

Eight K&J D42 magnets total: dia 6.35 x 3.175 [L]

Pocket dia 6.65; adhesive allowance 0.15 radial [A]

Remote magnets: (±13,16); dock (±13,-26) [A]

Phone magnets: (±14,±14); verify attraction polarity [A]

Separate steel phone plate: 45 x 45 x 0.5 [A]

Case/phone envelope width 84, height 180, thickness 10 [A]

Phone bottom alignment Y=-35 ±5; mark on grip [A]

Antenna in dock: Y=-70 to -64.9 [A,G]

Nearest permitted phone metal Y=-40: 24.9 separation.

Battery-antenna planar separation: 18.4 nominal.

Chosen metal separation target 15; Ebyte gives no numeric value.

Phone/plate RF effects still require physical testing.

[L] <https://www.kjmagnetics.com/d42-neodymium-disc-magnet>

MAGNETIC SHUTTER REMOTE | ORIGINAL DESIGN R2

Actuators, wiring and assembly inspection schedule

Button caps: dia 5.8 in dia 6.2 lid holes; radial clearance 0.2 [A]

Cap flange dia 8 x 0.8; stem dia 1.6; TPU compliant construction [A]

Rest stem tip Z=11.2; nominal switch top Z=11.0; free gap 0.2 [A,E]

Switch stroke 0.2 +/-0.1; force 160 +/-50 gf [E]

Adjust printed stem to measured switch height and test overtravel; do not preload the switch.

SW1 fork inside width 1.7 around nominal 1.3 actuator; travel 1.6 [A,F]

Both states must seat fully without side-loading the switch.

Harness: manufacturer 100 +/-3 length, 26 AWG UL3302 [I]

Two wire reservations dia 1.2, centre spacing 1.3; route through W1 with insulated/smooth edge [A]

Plan route passes side lanes at X=+/-11 and Y=24, above the cell, away from bottom electronics.

Retain full lead length with smooth loops; verify bend radius ≥ 3 and strain relief on actual assembly.

Wire ends must not press on cell seals, PCB holes, magnets or exposed solder joints.

Critical checks before any fabrication release:

1. Resolve the USB-C supplier-library/clearance blocker; repeat footprint, routing and export checks.
2. Compare imported electronic models and the complete assembled envelope after J1 is finalized.
3. Review stencil and through-hole shell soldering with the assembler; no assembler feedback obtained.

Prototype-only acceptance steps (not passed by these drawings):

Pack polarity and protection; charger voltage/current and hot/cold stops; pad thermal error and pressure; actual component heights, battery swelling margin, cable abrasion, button/slider force and travel; magnet retention, docking/drop tests; docked/detached RF; native iPhone and Android camera behavior.

Dimension sources are listed on sheets 1-3 and in mechanical/dimensions.json.

Original enclosure and grip design. No JJC internal dimensions are asserted.